Sand Dune

Design, Modeling, and Validation of a
Background Oriented Schlieren System
for High Resolution Refractive Index Mapping
of Transparent Media

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Abstract

Sand Dune is a custom Background Oriented Schlieren (BOS) system for quantitative measurement of weak refractive disturbances in transparent media. The project combines a dedicated optical rig with a desktop application that handles image loading, windowed cross-correlation, vector validation, surface reconstruction, visualization, and export. This document describes the optical geometry and physical model, the numerical processing pipeline in detail, how to use the application, and the testing strategy used to validate the software during development.

Contents

1 Introduction

Background Oriented Schlieren converts transverse refractive index gradients into an image-registration problem. A structured background pattern is imaged through a transparent specimen, and local refractive-index gradients cause apparent displacements of the pattern in the recorded image. Comparing a reference image (no specimen) with a disturbed image (specimen present) and measuring those pixel shifts gives the gradient field of the refractive-index distribution. Integrating those gradients yields a relative surface map of either the refractive-index variation or the thickness variation of the sample, depending on which quantity is assumed known.

BOS is attractive for weak optical disturbances, the kind where interferometry would be impractical and shadowgraphy lacks quantitative resolution. However, the accuracy of the final result depends on the entire measurement chain: background pattern design, optical geometry, image quality, sub-pixel estimation, validation, and reconstruction. Sand Dune was built to address that full problem as a single coherent system rather than as a stand-alone post-processing utility.

This document is organized as follows. Section 2 describes the physical setup. Section 3 presents the BOS optical model and the scaling equations implemented in the code. Section 4 describes each stage of the processing pipeline in detail, including the cross-correlator, CMM sub-pixel solver, optional PID deformation correction, validation, and surface reconstruction. Section 5 documents the codebase organization and key classes. Section 6 explains how to use the application. Section 7 summarizes the validation strategy and presents the development-verification precision figures used to characterize the current implementation. Section 8 derives a minimum resolvable gradient for the current optical configuration and compares it to the expected measurement range. Section 9 discusses practical calibration. Section 10 lists known caveats in the current implementation.

2 System Design

2.1 Measurement Objective

Sand Dune targets weak to moderate refractive disturbances in transparent samples, a regime where displacements are often sub-pixel, smooth, and low in magnitude. This makes the measurement sensitive to small errors at every stage: optical alignment, background contrast, sub-pixel fitting, and reconstruction. The physical system was therefore designed around three practical constraints:

- stable optical alignment between camera, specimen, and background,
- repeatable sample positioning,
- a background pattern with sufficiently dense local texture for reliable sub-pixel cross-correlation.

2.2 Optical Layout

In the assembled rig, a camera with a fixed focal-length lens images a patterned background target. The specimen or flow field is placed between the background and the lens. The four geometric parameters that define this layout are Z_d, Z_a, f, and P_{px}. These quantities are introduced formally in Section 3.1; here it is sufficient to note that they correspond to measurable hardware properties and inter-component distances set during assembly and calibration.

The system is closely aligned with the BOS geometry described by Raffel [1] and with later work on weak-flow extraction using dense background texture and careful geometric design [6].

2.3 Physical Hardware

The hardware consists of a camera, a fixed focal-length imaging lens, a patterned background target, a rigid support structure, and a specimen region between the background and the lens. The region of interest may occupy only part of the image frame, so the software supports a circular mask that restricts the reconstruction domain to the sample area and excludes the holder and other irrelevant structure. A custom holder was designed so that the specimen position is repeatable and the masked reconstruction domain matches the useful optical aperture of the setup.

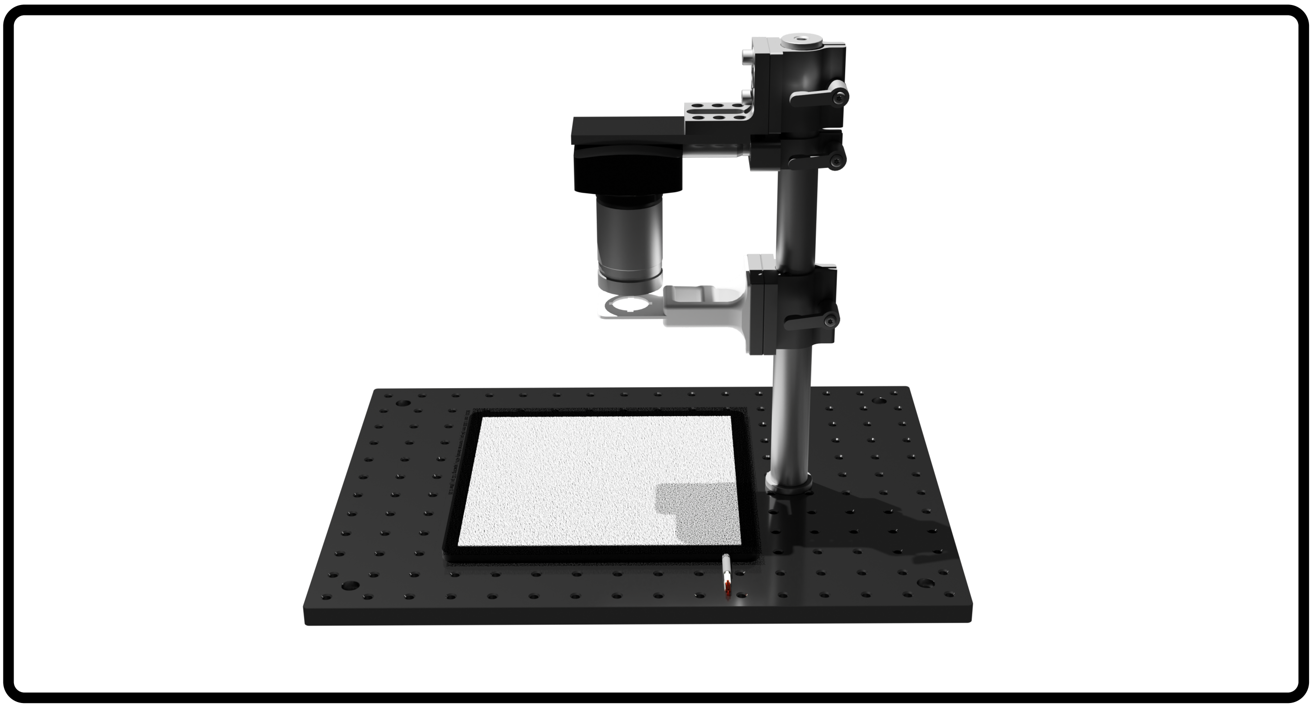

Figure 1: Hardware design of Sand Dune optical rig showing camera, lens, specimen region, and patterned background target.

2.3.1 Illumination

The background is illuminated by an LED backlight panel operated at full brightness. An incoherent light source is essential: coherent sources (lasers) produce speckle in the image plane, which adds structured noise to the cross-correlation and degrades sub-pixel accuracy. LEDs are suitable because they are spatially incoherent.

A blue optical filter is placed in the illumination path. White-light sources produce chromatic aberration, which manifests as colour fringing around high-contrast dot edges and introduces a small but systematic bias into the correlation displacement estimate. The blue filter narrows the effective spectral bandwidth, reducing chromatic aberration without requiring an apochromatic lens.

2.3.2 Background Patterns

Six printed background patterns were fabricated, covering two dot densities (40 % and 60 % fill) and three physical dot diameters (0.10 mm, 0.50 mm, and 0.75 mm). The pattern is chosen to match the imaging magnification so that each dot projects to approximately 4–5 pixels on the sensor. This size is the practical optimum for FFT-based cross-correlation:

- Dots smaller than ~ 3 px are under-sampled; the correlation peak is narrow and noisy, and the CMM bicubic fit has insufficient support to resolve sub-pixel offsets accurately.
- Dots larger than ~ 7 px reduce the number of independent features per interrogation window, lowering the signal-to-noise ratio of the correlation peak.
- At 4–5 px the peak is well-resolved, the CMM objective has a smooth, broad basin, and the S2N ratio is typically maximised.

Dot focus is equally important. An out-of-focus pattern blurs the dots to larger apparent diameters and reduces edge contrast. Both effects degrade the correlation peak shape, which CMM relies on for sub-pixel mapping. The background should be focused as sharply as the camera optics allow.

The 40 % density patterns provide more inter-dot spacing and are better suited to large interrogation windows or coarser magnification. The 60 % density patterns pack more texture per window and are preferred when small window sizes are used or when the dot diameter is near the lower end of the 4–5 px range. For the present optical configuration, the 0.10 mm, 40 % density pattern provides the most reliable correlation quality. For future configurations, the background should be re-selected after checking the projected dot size and the resulting correlation quality with the software.

2.3.3 *Imaging Lens and Upgrade Path*

The current setup uses a **25 mm fixed focal-length lens**. This provides a field of view and working distance compatible with the present sample size and mechanical layout. A longer focal-length lens remains a reasonable design option for future versions of the rig, but in BOS the consequences are best understood in terms of spatial specificity and the geometry required to preserve a usable field of view.

At fixed Z_a, increasing f enlarges the background image on the sensor, so each interrogation window corresponds to a smaller physical region in the sample plane. In BOS terms this makes the measurement more spatially specific: a localized refractive disturbance influences a narrower family of rays, which reduces lateral cross-talk between neighbouring regions and can make fine optical structure easier to isolate. The practical tradeoffs are:

- **Improved locality of the BOS response.** Because each pixel subtends a narrower bundle of rays at the sample plane, refractive features that are close together are less likely to contaminate the same interrogation windows. This can improve the lateral specificity of both the gradient fields and the reconstructed surface.
- **Background re-matching becomes necessary.** At the same stand-off distance, the background dots project to more pixels, so the physical dot diameter usually has to be reduced to remain in the preferred 4–5 px range. A longer lens therefore normally requires a different background pattern even if the rest of the rig is unchanged.
- **Field coverage becomes the primary geometric constraint.** If the same sample area must remain visible, the camera commonly has to be moved farther from the specimen and background. That change modifies Z_a, Z_B, and the sample-plane grid spacing, so the final BOS sensitivity is determined by the full reconfigured geometry rather than by focal length alone.
- **Sensitivity and specificity no longer move together automatically.** A longer lens can improve locality of the measured disturbance while still forcing a larger stand-off distance to keep the full sample in view. Depending on how the rig is re-scaled, that can either help or partially offset the increase in measurable pixel displacement.
- **Focus tolerance becomes tighter.** Longer lenses have shallower depth of field at a given aperture and working distance, so the background pattern and specimen surfaces must be positioned more carefully to avoid differential blur across the optical path.
- **Lens quality matters more as magnification increases.** Any substitution should be evaluated experimentally for contrast, aberration, and repeatability with the intended background pattern, because those effects directly influence the correlation peak shape used by the sub-pixel solver.

3 BOS Measurement Model

3.1 Optical Parameters

The software uses the following optical parameters, all configurable through the **Parameters** panel:

Symbol	Parameter name in UI	Description
P_{px}	Pixel pitch	Sensor pixel pitch (μm/pixel)
f	Focal length	Lens focal length (mm)
Z_d	Z_d	Background-to-sample distance (mm)
Z_a	Z_a	Sample-to-lens front principal plane distance (mm)
Z_B	N/A	$Z_B = Z_a + Z_d$, background-to-lens distance (mm)
t	Thickness	Sample thickness (mm)
n	Index	Sample refractive index (dimensionless)

The image distance for the background plane follows the thin-lens relation:

$$z_i = \frac{f Z_B}{Z_B - f}. \tag{1}$$

The lateral magnification of the background plane onto the sensor is

$$m_1 = \frac{z_i}{Z_B}. \tag{2}$$

3.2 What BOS Actually Measures

BOS fundamentally measures the gradient of the *optical path length* (OPL) through the sample, not the refractive index or the thickness independently. The optical path length through a sample at lateral position (x, y) is

$$\mathrm{OPL}(x, y) = \int_0^t n(x, y, z)\, dz, \tag{3}$$

where the integral runs along the optical axis through the full sample depth. For a sample that is thin relative to Z_a and optically homogeneous in depth but varying laterally, this reduces to $\mathrm{OPL}(x, y) \approx n(x, y)\, t$.

A lateral OPL gradient bends the transmitted ray. In the present software, that effect is modeled with a thin-slab, weak-deflection approximation in which the local angular deflection is proportional to the transverse refractive-index gradient:

$$\varepsilon_x \approx \frac{t}{n} \frac{\partial n}{\partial x}, \tag{4}$$

with the same expression used in the implemented scaling law. This model assumes a sample of approximately uniform thickness, weak angular deflection, and negligible variation through the optical depth. This deflected ray appears to originate from a laterally shifted background position. The apparent shift in the background plane is

$$\Delta x_{\mathrm{bg}} = \varepsilon_x\, Z_d, \tag{5}$$

and after projection through the imaging optics the corresponding image-plane displacement in physical units is

$$\Delta x_{\mathrm{img}} = \varepsilon_x\, Z_d\, \frac{z_i}{Z_B}. \tag{6}$$

In pixels, $u_{px} = \Delta x_{\mathrm{img}} / P_{px}$.

Decoupling limitation. Because BOS measures OPL, it cannot independently separate $n(x, y)$ from $t(x, y)$ without additional information. The two reconstruction modes in the software address this by assuming one quantity is known:

- **RI Variation (dn mode):** the sample thickness t is assumed known and spatially uniform; the software reconstructs a relative refractive-index field $\Delta n(x, y)$.
- **Thickness Variation (mm mode):** the sample refractive index n is assumed known and spatially uniform; the software reconstructs thickness variation $\Delta t(x, y)$.

Both modes are simplifications of the more general OPL measurement. Any variation in the quantity assumed to be fixed will appear as a spurious contribution to the reconstructed field.

Other physical contributions. Several effects beyond the OPL gradient influence the measured displacement and should be considered when interpreting results:

- **Refraction at flat surfaces:** a ray entering a flat-faced sample at an angle undergoes lateral displacement by Snell's law even with no internal RI gradient. This geometric offset scales with sample thickness and off-axis angle, and is addressed by the optional field correction (Section 3.6).
- **Sample tilt:** if the sample faces are not perpendicular to the optical axis, a systematic displacement offset is introduced across the entire field. This appears as a planar background in the u and v maps and in the reconstructed surface.
- **Defocus of background texture:** the effective background contrast changes slightly with RI variations, because the sample acts as a weak lens. In practice this effect is very small for weak disturbances.

3.3 Displacement-to-Gradient Scaling

The conversion from a raw pixel displacement to a physical gradient is implemented in `Session::ScaleFields()`. Combining Equations (4) and (6) gives

$$u_{px} = \frac{\Delta x_{\text{img}}}{P_{px}} = \frac{t}{n} \frac{\partial n}{\partial x} \frac{Z_d}{P_{px}} \frac{z_i}{Z_B}, \tag{7}$$

where, for readability in the equations below, P_{px} is interpreted in mm/pixel. In the UI and code implementation the pixel pitch is entered in µm/pixel, so `Session::ScaleFields()` multiplies it by 10^{-3} before applying the same formulas.

Using

$$\frac{z_i}{Z_B} = \frac{f}{Z_B - f}, \tag{8}$$

the RI-variation inversion becomes

$$\frac{\partial n}{\partial x} = u_{px} P_{px} \frac{n (Z_B - f)}{f Z_d t}. \tag{9}$$

The code therefore applies the correlation-map scale factor

$$S_{\text{corr}} = P_{px} \frac{n (Z_B - f)}{f Z_d t}, \tag{10}$$

so that

$$\frac{\partial n}{\partial x} = u_{px} S_{\text{corr}}. \tag{11}$$

For thickness-variation mode, the implemented model replaces t with $(n - 1)$:

$$S_{\text{corr}} = P_{px} \frac{n (Z_B - f)}{f Z_d (n - 1)}. \tag{12}$$

3.4 Sample-Plane Grid Spacing

The displacement field is sampled on a correlation grid rather than on individual pixels. The grid step (in image pixels) is

$$\texttt{step} = \max(1,\ W - O), \tag{13}$$

where W is the window size and O is the overlap. The corresponding physical spacing in the sample plane is

$$\Delta x_{\text{grid}} = \texttt{step} \cdot P_{px} \cdot \frac{Z_a}{z_i}. \tag{14}$$

3.5 Surface Scale Factor

Because reconstruction integrates gradient values across discrete grid cells, the reconstructed surface values must also absorb the physical cell width. Therefore the surface scale factor is

$$S_{\text{surf}} = S_{\text{corr}} \times \Delta x_{\text{grid}}. \tag{15}$$

Both the displayed physical axis width and the reconstructed surface amplitude therefore depend on the same grid-spacing factor. Correcting the geometric parameters to fix the displayed axis also changes the reconstructed surface amplitude.

3.6 Optional Refraction Correction

For samples with significant thickness, the curvature of off-axis rays through the slab produces a background-plane displacement even for a spectrally uniform sample. The software can compute and subtract this geometric artifact via `Session::ComputeRefractionCorrection()`.

For each correlator-grid point at column c, row r, the correction is derived from Snell's law:

$$\theta_x = \arctan\left(\frac{r_x}{Z_a}\right), \qquad \theta_y = \arctan\left(\frac{r_y}{Z_a}\right), \tag{16}$$

$$\theta_{r,x} = \arcsin\left(\frac{\sin\theta_x}{n}\right), \qquad \theta_{r,y} = \arcsin\left(\frac{\sin\theta_y}{n}\right), \tag{17}$$

where

$$r_x = \left(c - \frac{w}{2}\right)\ \texttt{step}\ P_{px}\,\frac{Z_a}{z_i} \tag{18}$$

is the sample-plane lateral position, with an analogous expression for r_y.

The lateral displacement of the ray inside the slab is

$$d_x = \frac{\sin(\theta_x - \theta_{r,x})}{\cos\theta_{r,x}}\,t, \tag{19}$$

and similarly for d_y. Projecting back to the sensor plane and converting to pixels gives the correction field that is subtracted before any further scaling.

This correction should be considered an optional preprocessing step. Its importance scales with both sample thickness and the departure of n from unity: for samples with refractive indices close to 1, the Snell's-law offset is very small and the correction has negligible impact on the final result. It becomes progressively more important as n increases or as t grows large.

4 Processing Pipeline

Figure 2 shows the three-stage pipeline.

(Pipeline diagram can be placed here.)

Figure 2: Overview of the Sand Dune processing pipeline.

4.1 Stage 1: Cross-Correlation

4.1.1 *Image Preparation*

Images are loaded as normalized floating-point matrices by the `Image` class. Pixel values are mapped to $[0, 1]$. The correlator operates directly on these Eigen matrices without further preprocessing.

4.1.2 *Window Grid*

The image is divided into a regular grid of overlapping interrogation windows. For a window size W and overlap O, the grid step is given by Equation (13). The number of grid columns and rows is

$$n_c = \left\lfloor \frac{W_{\text{img}} - W}{\texttt{step}} \right\rfloor + 1, \qquad n_r = \left\lfloor \frac{H_{\text{img}} - W}{\texttt{step}} \right\rfloor + 1. \tag{20}$$

4.1.3 *FFT Cross-Correlation*

For each window pair (reference and flow), the correlator:

1. extracts the two interrogation blocks,
2. skips the window if the local variance is below 10^{-3} (flat patch),
3. subtracts the local mean from each window,
4. computes the forward FFT of both windows,
5. forms the cross-power spectrum $G = \overline{F_{\text{ref}}}\, F_{\text{flow}}$,
6. applies the inverse FFT to obtain the raw correlation map,
7. shifts the map so that zero displacement lies at the center.

The resulting map ϕ satisfies

$$\phi = \mathcal{F}^{-1}\left\{\overline{F_{\text{ref}}}\, F_{\text{flow}}\right\}. \tag{21}$$

The signal-to-noise ratio is defined as

$$\text{s2n} = \frac{\phi_1}{\phi_2}, \tag{22}$$

where ϕ_1 is the primary peak value and ϕ_2 is the highest value outside a ± 2-pixel exclusion zone around the primary peak.

4.1.4 *CMM Sub-Pixel Refinement*

Sub-pixel refinement is implemented using the Correlation Mapping Method (CMM) of Chen and Katz [2]. The solver operates on a 5×5 patch centred on the discrete peak.

The objective is to minimize the mean-normalized patch residual

$$\varepsilon = \sum_{i=0}^{24} \left(\tilde{\phi}[i] - \tilde{\phi}'[i]\right)^2, \tag{23}$$

where $\tilde{\phi}[i] = \phi[i]/\bar{\phi}$ is the measured patch normalized by its mean, and $\tilde{\phi}'[i]$ is the autocorrelation patch evaluated at the sub-pixel offset (u', v') via bicubic interpolation. The autocorrelation patch is built once per window from the saved reference FFT.

Minimization proceeds as follows:

1. **Seed:** a log-parabola fit along the dominant axis gives an initial (u'_0, v'_0).
2. **Gauss–Newton:** up to 8 iterations with backtracking line search (step sizes $\{1, 0.5, 0.25, 0.1, 0.05\}$). The Jacobian uses the analytic bicubic derivative.

3. **Grid fallback:** if the Gauss–Newton solution does not improve the seed, a dense grid search on a 0.01-pixel grid is used.
4. **Boundary handoff:** if the solution is within 0.02 px of ± 0.5, the adjacent integer cell is tested and the lower-residual representation is kept.

The solution is clamped to $|u'| \leq 0.49$, $|v'| \leq 0.49$ to prevent ambiguous half-pixel values.

Peak locking. A key motivation for using CMM rather than a simpler centroid or parabolic fit is the suppression of peak locking. Conventional sub-pixel estimators that fit a symmetric function to the discrete correlation peak are biased toward integer and half-integer values, producing characteristic spikes in the displacement histogram. CMM avoids this by matching the entire local patch shape to a reference autocorrelation rather than fitting only the peak tip. The boundary-handoff logic and the clamp at ± 0.49 px are additional safeguards that prevent the estimate from locking to cell boundaries. The peak-locking fraction is tracked explicitly in the test suite (Section 7) to confirm that this behaviour remains controlled.

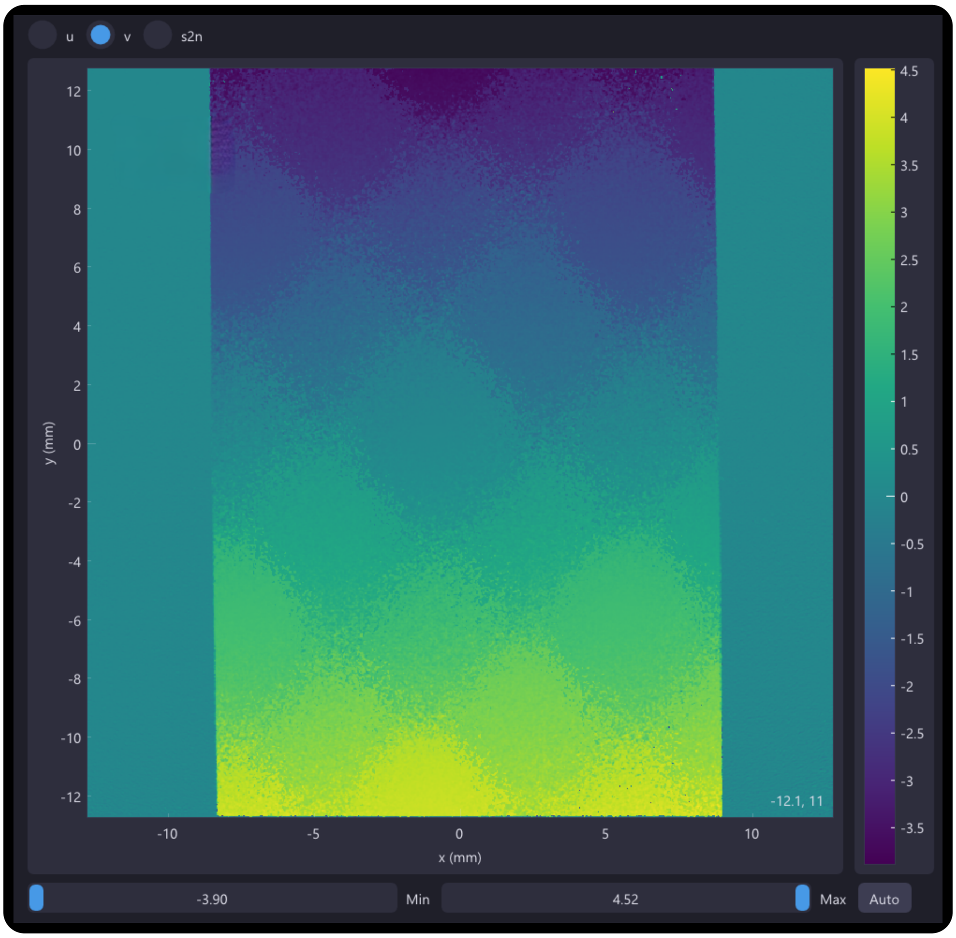

Figure 3: An example of peak locking patterns formed from a soft gradient measurement field.

4.1.5 *Progressive Image Deformation (Optional)*

When `enable_pid` is active, the correlator performs Progressive Image Deformation (PID) to account for local shear and strain in the displacement field. PID iteratively warps the interrogation windows to match the local deformation model, so that each correlation pass measures only the residual displacement after the current best estimate is removed.

Algorithm. Starting from the baseline rigid-window correlation:

1. Compute first-order displacement gradients $\partial u/\partial x$, $\partial u/\partial y$, $\partial v/\partial x$, $\partial v/\partial y$ via central differences at the correlator-grid spacing.
2. Apply `pid_smoothing_passes` passes of a 3×3 box filter to the deformation field.
3. For each window, construct an affine warp: $\mathbf{x}_{\text{warped}} = \mathbf{x} + \mathbf{u}(\mathbf{x}_c) + J\,(\mathbf{x} - \mathbf{x}_c)$, where $\mathbf{x}_c$ is the window centre and J is the local deformation Jacobian.
4. Resample the flow window using bilinear interpolation and re-correlate to measure the residual displacement.
5. Accept the update for a window only if the CMM residual improves (or the S2N improves when residuals are not available).
6. Repeat for `pid_iterations` additional passes. If the mean correction across all windows falls below 10^{-3} px, terminate early.

The `pid_relaxation` factor damps the accepted residual updates to improve convergence stability.

4.2 Stage 2: Validation

Validation is implemented in `src/grains/validation.cpp` and uses a two-pass outlier-detection strategy.

Pass 1: Signal-to-noise threshold. A vector is marked invalid if

$$\text{s2n} < 1.3. \tag{24}$$

Pass 2: Normalized residual (UNO criterion). For each vector the 8-connected neighbourhood median is computed:

$$u_{\text{med}} = \operatorname{median}\big(u_k,\ k \in \mathcal{N}_8\big). \tag{25}$$

The normalized deviation is

$$\tilde{u}_{\text{nrm}} = \frac{|u - u_{\text{med}}|}{\operatorname{median}(|u_k - u_{\text{med}}|) + \varepsilon}, \qquad \varepsilon = 0.1, \tag{26}$$

and the vector is flagged if

$$\sqrt{\tilde{u}_{\text{nrm}}^2 + \tilde{v}_{\text{nrm}}^2} > 2.0. \tag{27}$$

This is the universal outlier detection (UNO) criterion of Westerweel and Scarano [3].

Replacement. All flagged vectors are replaced with the 8-neighbourhood median of the unflagged neighbours.

4.3 Stage 3: Surface Reconstruction

4.3.1 *Masked Sparse Poisson Solver (Default Method)*

The default reconstruction method builds and solves a sparse linear system on the masked correlation grid. Let $s_{i,j}$ be the unknown scalar field on valid nodes. Edge gradients are formed as

$$g^x_{i,j} = \tfrac{1}{2}\left(u_{i,j} + u_{i,j+1}\right), \tag{28}$$

$$g^y_{i,j} = \tfrac{1}{2}\left(v_{i,j} + v_{i+1,j}\right), \tag{29}$$

and the energy to be minimized is

$$E(s) = \sum_{\text{x-edges}} \left(s_{i,j+1} - s_{i,j} - g^x_{i,j}\right)^2 + \sum_{\text{y-edges}} \left(s_{i+1,j} - s_{i,j} - g^y_{i,j}\right)^2. \tag{30}$$

Setting $\partial E/\partial s_{i,j} = 0$ yields a standard sparse Poisson-style linear system. The domain is restricted to nodes inside the mask, so no artificial boundary data is inserted. The system is assembled with Eigen triplets and solved using `Eigen::ConjugateGradient` with tolerance 10^{-5} and a maximum of $\max(4N, 200)$ iterations, where N is the number of unknowns.

The constant-offset nullspace is resolved by a soft gauge condition on the first valid interior node (pinned to zero). The reconstructed field is therefore relative in absolute value but meaningful in shape.

4.3.2 *Frankot–Chellappa (Optional Method)*

The software also provides an FFT-based reconstruction option following Frankot and Chellappa [4]. This path is exposed in the UI as an alternative integrator and is implemented by `ComputeFC()`. It mirrors the displacement field into a 2×2-tiled domain to enforce periodic boundary conditions and then integrates in the frequency domain:

$$\hat{s}(f_x, f_y) = \frac{i f_x\, \hat{u}(f_x, f_y) + i f_y\, \hat{v}(f_x, f_y)}{2\pi(f_x^2 + f_y^2) + \varepsilon}. \tag{31}$$

On full-domain inputs, the Frankot–Chellappa method can produce results comparable to the sparse Poisson solver and is substantially faster, since it requires only a small number of FFTs regardless of grid size. Its main caveat in the present BOS workflow is the treatment of the sample mask. The current implementation applies the mask by zeroing u and v outside the valid region before the FFT integration. This is convenient and keeps the option usable in the UI, but it does not treat the exterior as an unknown domain in the same way as the sparse solver. As a result, sharp mask boundaries can introduce artificial low-frequency structure near the sample edge. For masked aerogel-style datasets, the sparse Poisson solver is therefore the default and generally the more trustworthy choice, while Frankot–Chellappa remains useful as a fast comparison method on cleaner or less heavily masked fields.

5 Software Architecture

5.1 Build System and Dependencies

The project is built with CMake (C++23 required). External dependencies are:

Library	Purpose
SDL3, SDL3_image	Application window, rendering, image file I/O
Dear ImGui	All UI panels and controls
ImPlot	Heatmap visualizations
Eigen3	Dense/sparse linear algebra
FFTW3 (float)	FFT cross-correlation and Frankot–Chellappa
doctest	Unit-test framework (test binary only)

The build produces two targets:

- `sand_dune`: the GUI application,
- `run_tests`: the standalone test binary.

5.2 Module Organization

Namespace / directory	Classes	Role
src/main.cpp		Application entry point
include/session.h, src/session.cpp	Session	Pipeline orchestration, scaling, async execution, CSV export
include/grains/	Correlator	FFT correlation, CMM, PID
src/grains/	Validation	Outlier detection & replacement
	Reconstruction	Poisson / Frankot–Chellappa integration
include/water/ src/water/	UI, App	ImGui panels, SDL3 shell
include/sun/ src/sun/	Image, Mask	Image loading, mask generation
include/wind/ src/wind/	VectorField (parameter structs)	Displacement-field container
tests/		Development verification tests

5.3 Session: Pipeline Orchestrator

`Session` (`include/session.h`) is the top-level controller. It owns the input images, all intermediate data arrays, the parameter structs, and the stage-state machine:

```
// Images
Image ref;
std::vector<Image> flows;

// Intermediate data
std::vector<VectorField> raw_correlation_field;
std::vector<VectorField> correlation_field;
std::vector<VectorField> raw_val_field;
std::vector<VectorField> val_field;
std::vector<Eigen::MatrixXf> raw_surface;
std::vector<Eigen::MatrixXf> surface;
Eigen::MatrixXf correction[2]; // refraction correction (u, v)

// Parameters
CorrelatorParameters correlatorparameters;
OpticalParameters    opticalparameters;
Mask                 mask;

// Stage state (per stage: Idle, Ready, Busy, Done, Dirty)
std::atomic<StageState> stagestates[STAGE_TOTAL];
```

The three pipeline stages run asynchronously via `std::async`. The UI polls the stage states to display progress and enables/disables run buttons accordingly.

5.4 Correlator

`Correlator` (`include/grains/correlator.h`) takes a reference `Image` and a flow `Image` and returns a `VectorField`. Internally it allocates FFTW plans once at construction time and reuses the same transform buffers for every window.

```
struct CorrelatorParameters {
    int    window_size        = 32;
```

```
    int   overlap            = 24;
    bool  enable_pid         = false;
    int   pid_iterations     = 2;
    float pid_relaxation     = 1.0f;
    int   pid_smoothing_passes = 1;
};
```

The main entry point is `Compute(ref, flow)`, which dispatches to `ComputeWithPid()` or `ComputeSinglePass()` based on `enable_pid`.

5.5 Reconstruction

`Reconstruction` (`include/grains/reconstruction.h`) provides two integration back ends. The default path takes a `VectorField` and a binary mask `Eigen::MatrixXi`, builds an index map that assigns consecutive unknowns only to valid interior cells, constructs the sparse system from edge-gradient equations, and calls the Eigen conjugate-gradient solver. An FFT-based Frankot–Chellappa path is also available as an alternative reconstruction option.

5.6 VectorField

`VectorField` (`include/wind/vectorfield.h`) stores three `Eigen::MatrixXf` matrices (u, v, s2n) and grid dimensions. It exposes a `SaveCSV(path)` method that writes columns `col`, `row`, `u`, `v`, `s2n` in column-major order.

6 Using the Application

6.1 Typical Workflow

1. **Load images.** Use the **Load** panel to browse for a reference image and one or more disturbed (flow) images. All images must share the same pixel dimensions.

2. **Set parameters.** In the **Parameters** panel, enter the optical geometry (Z_d, Z_a, f, P_{px}), sample properties (n, t), and the correlation settings (window size, overlap, PID toggle). The **Calculations** panel updates in real time to show the derived grid spacing, physical field width, and estimated feature size; use these to check that the window size is sensible for the expected background texture.

3. **Configure the mask.** In the **Parameters** panel, enable the circular mask and set its centre and radius to enclose the sample region. Reconstruction will be confined to this domain.

4. **Run the pipeline.** In the **Pipeline** panel, click **Run Full Pipeline** to run all three stages sequentially, or run each stage individually. A progress bar and ETA are shown for each stage. With multiple flow images, all frames are processed in the same batch.

5. **Inspect results.** Switch between the **Correlation**, **Validation**, and **Surface** views using the tabs in the visualization panel. For correlation and validation, select the u, v, or s2n map, and toggle between raw pixel units and scaled physical units. Adjust the colormap range using the min/max sliders.

6. **Export.** In the **Save** panel, choose an output directory and click **Save**. Three CSV files are written: `_correlation.csv`, `_val.csv`, and `_surface.csv`. For multi-frame batches, a numeric suffix is appended per frame.

6.2 Panel Reference

6.2.1 Load Panel

Provides file-picker buttons to select the reference image and one or more flow images. The flow image list supports multi-frame batches. When multiple flow images are loaded, navigation arrows are shown in the visualization panels to step through frames.

6.2.2 Parameters Panel

Group	Control	Effect
Correlation	Window size	Interrogation window side length (pixels)
	Overlap	Window overlap (pixels; step = size − overlap)
	Enable PID	Enables iterative window deformation
	PID iterations	Number of PID passes after baseline
	PID relaxation	Damping on residual updates ($[0, 1]$)
	PID smoothing	3×3 smoothing passes on deformation field
Optical	P_{px}	Sensor pixel pitch (µm)
	Z_d	Background-to-sample distance (mm)
	Z_a	Sample-to-lens distance (mm)
	f	Lens focal length (mm)
Sample	n	Sample refractive index
	t	Sample thickness (mm)
Reconstruction	Mode	RI variation (dn) or thickness variation (mm)
	Apply correction	Subtract Snell's-law refraction offset
Mask	Enable mask	Restrict reconstruction to circular ROI
	Centre (x, y)	Mask centre in correlation-grid coordinates
	Radius	Mask radius in correlation-grid cells

6.2.3 Calculations Panel

Displays derived quantities updated in real time from the current parameters: correlation-grid step (px), sample-plane grid spacing (mm), physical field width (mm), image magnification, and the expected pixel displacement for a reference gradient magnitude.

6.2.4 Pipeline Panel

Shows three stage rows (Correlation, Validation, Reconstruction) each with a **Run** button, a progress bar, and a status label (Idle / Ready / Busy / Done / Dirty). A **Run Full Pipeline** button chains all three stages.

6.2.5 Visualization Panel

- **Correlation / Validation tabs:** heatmap of the u, v, or s2n map. A toggle switches between raw pixels and scaled physical units. Min/max sliders control the colormap range.
- **Surface tab:** heatmap of the reconstructed scalar field. Axes are labelled in physical millimetres. Min/max sliders control the colormap range. A tooltip shows the value under the cursor.

6.2.6 *Save Panel*

Allows choosing an output directory (via system file picker) and exports the current results to CSV. Saving runs asynchronously to avoid blocking the UI.

6.2.7 *Settings Panel*

Offers a dark/light theme toggle.

6.3 CSV File Formats

Correlation and Validation CSV. The correlation and validation result files share the following layout:

```
rows, cols
<height>, <width>
u, v, s2n
<u00>, <v00>, <s2n00>
...
```

Values are in physical units when optical parameters have been set; otherwise they are in raw pixels.

Surface CSV. The surface file uses column-major ordering:

```
rows, cols
<height>, <width>
<z[0,0]>
<z[1,0]>
...
```

Values are stored in column-major order (all rows of column 0, then all rows of column 1, etc.) in the units determined by the selected reconstruction mode (dn or mm).

7 Validation, Precision, and Repeatability

7.1 Overview

Validation is performed at two levels. At the vector level, the pipeline uses signal-to-noise thresholding and neighbourhood-median outlier detection (Section 4.2) to clean each displacement field in-line. At the software level, the development verification suite in `tests/test.cpp` was used to characterize numerical behaviour, estimate precision, and check for artifacts during implementation. The test binary (`run_tests`) is not part of the normal user workflow; it exists as a development and regression tool.

7.2 Image-Based Precision Metric

The most important quantitative precision figure available for the current system comes from the measured image pair `if_0.1`, stored under `images/slides`. This image pair was taken of a glass optical window of known uniform refractive index and thickness. The expected displacement field is smooth and nearly planar, so the residual from a best-fit plane directly quantifies spurious structure such as striping, local roughness, and sub-pixel locking artifacts.

During software verification, the test *Correlation Repo Image Pair* fits a least-squares plane to the recovered u and v fields and reports the RMS deviation. Results from the current codebase are summarised below.

Metric	u (px)	v (px)
Whole-map RMS, no PID	0.659 215	0.669 790
Whole-map RMS, PID	0.595 992	0.579 117
Central-crop RMS, no PID	0.096 441	0.105 765
Central-crop RMS, PID	0.048 692	0.045 640

The central crop removes the outer 10 % border on each side (containing ringing and artifacts from the sample holder and image edge), leaving the central 80 % of the map, which is the most representative region for the present system. Enabling PID reduces the central-crop RMS by approximately **49.5 %** in u and **56.8 %** in v.

7.3 Artifact Diagnostics

The development verification suite also includes image-based diagnostics designed to identify structured artifacts in smooth gradient fields:

- **Striping check:** the cropped displacement field is compared against its expected gradient axis using a normalized correlation coefficient. For the `if_0.1` image pair, the raw u and v fields give dominant-axis correlations of 0.9985 and 0.9981, while the orthogonal-axis correlations are only 0.0052 and 0.0068. After validation, these become 0.9985 and 0.9982 on the dominant axis, with orthogonal-axis correlations of 0.0052 and 0.0067. These values show a strongly one-dimensional gradient with no visible cross-axis striping in the central cropped region.
- **Peak-locking check:** the fraction of recovered displacement values locked near the half-pixel bands is measured together with the size of the largest connected locked region. For `if_0.1`, the measured locked fractions are 0.0430 and 0.0436 for the raw u and v fields, and 0.0425 and 0.0431 after validation. The largest connected locked regions contain 21 cells in u and 27 cells in v, both before and after validation, indicating that any peak-locking is weak and spatially limited. Additionally, some gradient points may be close to peak values and contributing to this metric inadvertently.
- **Validated residual ratio:** the ratio of the validation-stage residual to the correlation residual is tracked to confirm that validation does not hide structured error. For `if_0.1`, the residual ratio decreases from 0.0517 to 0.0505 in u and from 0.0554 to 0.0538 in v, showing a small cleanup of the field without altering its dominant gradient structure.

7.4 Supplementary Software Checks

In addition to the measured-image verification above, the *Correlator Computation* test case in `tests/test.cpp` was used during development to verify expected behaviour on controlled synthetic inputs. These checks are useful regression guards, but they are not the primary experimental precision metric for the present system.

Zero displacement. A random-noise image is correlated against itself. The measured RMSE against the zero-displacement ground truth is 0.0000 px, with a maximum absolute displacement of 0.0000 px anywhere on the grid. This confirms that the correlator introduces no systematic bias on a flat-spectrum input.

Sub-pixel translation. A synthetic spot-field (120 Gaussian spots, $\sigma = 1.1$ px, rendered on a 192×192 grid) is shifted by bilinear interpolation for four imposed offsets and the per-shift RMSE against the known ground-truth displacement is reported:

Imposed shift (px)	RMSE (px)
$(+0.25,\ -0.30)$	0.0633
$(+0.10,\ +0.12)$	0.0254
$(-0.35,\ +0.28)$	0.0711
$(+0.42,\ -0.18)$	0.0714

All four cases remain well below the 0.08 px acceptance threshold, confirming that the CMM sub-pixel solver operates correctly across the ± 0.5 px cell.

PID affine deformation. A spot-field (160 spots, same parameters) is warped by an affine transformation with both translation and first-order shear: $u_0 = 1.15$ px, $v_0 = -0.75$ px, $\partial u/\partial x = 0.0105$, $\partial u/\partial y = 0.0040$, $\partial v/\partial x = -0.0035$, $\partial v/\partial y = 0.0085$. The RMSE against the analytical displacement field is compared across iteration counts:

Method	RMSE (px)
Baseline (no PID)	0.2745
PID (2 iterations)	0.0875
PID (3 iterations)	0.0846

Two PID iterations reduce the RMSE by 68.1 %, from 0.2745 px to 0.0875 px. The residual is sub-pixel and consistent with the correlation noise floor, indicating that the deformation correction removes systematic error rather than merely redistributing it. A third iteration yields only a further 3.3 % reduction (0.0846 px), demonstrating the expected diminishing returns: most of the correctable deformation error is captured in the first two passes.

7.5 Repeatability

Repeatability is evaluated experimentally by acquiring repeated image sets under different distance and rotation conditions to asses the measured field:

- **Rotation symmetry:** the repeated acquisition of the sample was done slightly rotated to assess overall gradient symmetry and mapping quality.
- **Distance symmetry:** acquisitions at controlled Z_A distances were done to assess optical parameter impact and repeatability between such changes.

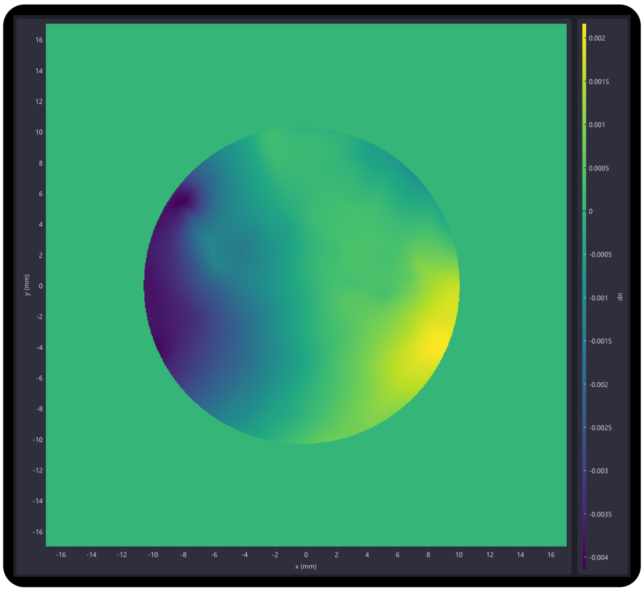

Figure 4: Refractive index map for a further placement from sample to lens.

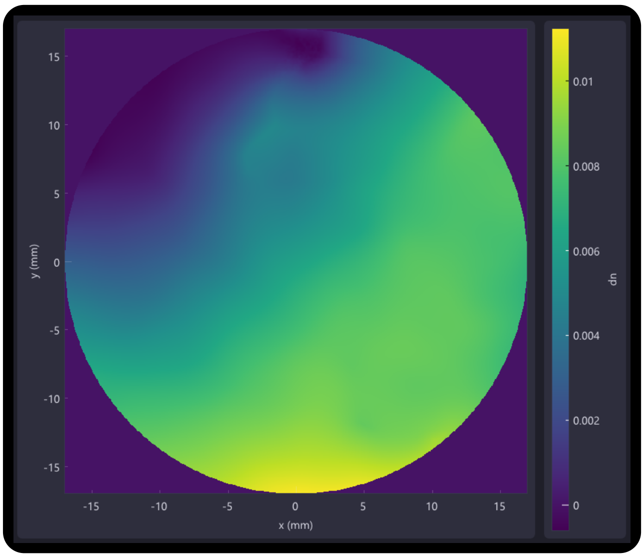

Figure 5: Refractive index map for a near placement of the same sample.

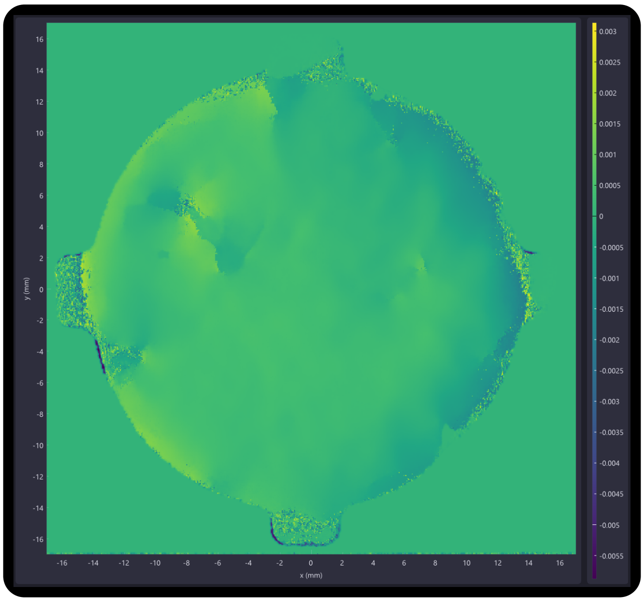

Figure 6: Refractive index gradient map for a further placement from sample to lens.

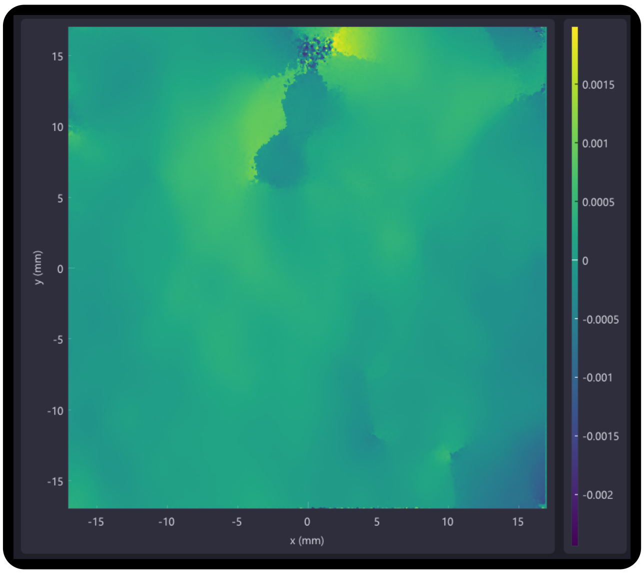

Figure 7: Refractive index gradient map for a near placement of the same sample.

The four figures above showed good repeatability and consistency between both changing setup geometries and reference image changes, as each used difference reference patterns for correlation and measurement.

8 System Precision Estimate

This section quantifies the minimum resolvable refractive-index gradient for the current optical configuration and places it in the context of the weakest gradients expected in practice.

8.1 Current Optical Setup

The figures below are computed for the following configuration:

Parameter	Symbol	Value
Lens focal length	f	25 mm
Background–sample dist.	Z_d	220 mm
Sample–lens distance	Z_a	45 mm
Total object distance	Z_B	265 mm
Pixel pitch	P_{px}	2.315 µm
Sample refractive index	n	1.08
Sample thickness	t	0.924 mm

8.2 Minimum Resolvable Gradient

The minimum detectable displacement is taken as three times the PID central-crop plane-fit RMS from the slide validation tests (Section 7):

$$u_{\min} = 3 \times 0.049 \approx 0.15 \text{ px}. \tag{32}$$

Using the three-sigma criterion gives a conservative threshold that distinguishes genuine signal from measurement noise at high confidence.

The displacement-to-gradient scale factor (Equation (10)), evaluated in SI units throughout, is

$$S_{\text{corr}} = \frac{P_{px}\,(Z_B - f)\,n}{f\,Z_d\,t} = \frac{(2.315 \times 10^{-6})(240 \times 10^{-3})(1.08)}{(25 \times 10^{-3})(220 \times 10^{-3})(0.924 \times 10^{-3})} \approx 0.1181 \text{ m}^{-1}\,\text{px}^{-1}. \tag{33}$$

The minimum resolvable refractive-index gradient is therefore

$$\left.\frac{dn}{dx}\right|_{\min} = S_{\text{corr}} \times u_{\min} = 0.1181 \times 0.15 \approx 1.77 \times 10^{-2} \text{ m}^{-1} = 1.77 \times 10^{-5} \text{ dn/mm}. \tag{34}$$

8.3 Expected Worst-Case Gradient

As a benchmark, consider a transparent sample with a total refractive-index variation of $\Delta n = 0.005$ distributed uniformly across the full sample diameter of 24 mm. This represents an internally measured contrast value assumed to be spread as a slow, gradual gradient over the entire aperture, which constitutes the worst case in the sense that the gradient is as shallow as it can be for a given total RI excursion:

$$\left.\frac{dn}{dx}\right|_{\text{wc}} = \frac{\Delta n}{d} = \frac{0.005}{24} \approx 2.08 \times 10^{-4} \text{ dn/mm}. \tag{35}$$

8.4 Detection Margin

The ratio of the expected worst-case gradient to the minimum resolvable gradient gives the system margin:

$$\text{margin} = \frac{(dn/dx)_{\text{wc}}}{(dn/dx)_{\min}} = \frac{2.08 \times 10^{-4}}{1.77 \times 10^{-5}} \approx 11.8. \tag{36}$$

Even in the worst case of a maximally shallow gradient, the current system can resolve the expected RI variation by approximately one order of magnitude. Steeper local gradients (more concentrated RI variations) would be resolved with even greater margin. This confirms that the optical sensitivity of the current setup is well matched to the target measurement regime.

9 Calibration and Interpretation

The most sensitive practical issue is geometric calibration, specifically setting Z_a correctly. In the software model, Z_a is an effective optical distance, not simply a mechanical measurement from the lens barrel. An incorrect Z_a shifts both the displayed field width and the reconstructed surface amplitude simultaneously, because both depend on the same grid-spacing factor Δx_{grid} (Equation (14)).

The recommended calibration procedure is therefore not a direct ruler measurement. Instead:

1. Place a target of known physical width in the background plane.
2. Record how many pixels span that width.
3. Infer the effective magnification and back-calculate Z_a so that the reported field width in the **Calculations** panel matches the known dimension.

Once Z_a is correctly calibrated, the surface amplitude will also be on the correct absolute scale.

The reconstructed surface is always relative in absolute value (the gauge constant is arbitrary), so absolute-offset comparisons between separate experiments require an independent reference point.

9.1 Interpreting Results: Reconstruction vs. Gradient Maps

The reconstruction and the raw displacement maps each reveal different aspects of the optical quality of a transparent sample, and both should be examined together rather than relying on either alone.

The reconstructed surface (dn or thickness map) gives a global view of the spatial distribution of optical path length across the sample. Broad, smooth features such as overall refractive-index gradients, wedge-shaped thickness variation, or large-scale density non-uniformities appear clearly in the integrated field. This view is well suited to comparing samples, measuring the amplitude of a known disturbance, or checking that the reconstructed field has the expected large-scale shape.

The correlation u and v gradient maps, by contrast, directly show where harsh or rapid variations in refractive index occur within the sample. Sharp features such as inclusions, bubbles, cracks, or regions of steep RI gradient produce strong, localised displacement peaks in these maps that may be smeared or suppressed by the spatial-integration step of the reconstruction. Examining the gradient maps in addition to the surface is therefore important for identifying optically defective regions that a smooth reconstruction might underrepresent.

In practice: use the reconstructed surface to characterise the overall optical uniformity and quantify the amplitude of large-scale variations; use the u and v maps to locate and assess localised optical defects.

10 Known Caveats

10.1 Fundamental Measurement Limitations

- **Refractive index and thickness cannot be decoupled.** BOS measures the integrated optical path length (OPL) through the sample. A thicker region with uniform refractive index produces exactly the same deflection signal as a thinner region with a higher refractive index. Without an independent thickness measurement (e.g. profilometry, OCT), the two contributions cannot be separated from the BOS data alone.

- **The reconstructed map is not a pure refractive-index map.** Although the output is labelled in refractive-index units via the Gladstone-Dale scaling, the deflection signal contains contributions from all sources that bend the light ray:
  - refractive index variation within the sample bulk,

  - physical tilt or wedge of the sample surfaces (a flat but tilted window deflects light even with perfectly uniform RI),
  - surface curvature and figure errors,
  - internal stress birefringence (though BOS is not polarisation-resolved and therefore only captures the scalar mean),
  - any other optical inhomogeneity along the line of sight.

  All of these contribute to the measured gradient field and are integrated into the final surface by the reconstruction step. There is no way to attribute a given feature in the output map to one specific physical cause without additional independent measurements.

- **BOS is a line-of-sight-integrated measurement.** The technique collapses the full 3D refractive-index field onto a single 2D projection. Any variation along the optical axis (depth within the sample) is summed and cannot be resolved. Tomographic BOS can recover 3D structure but requires multiple viewing angles and is not implemented here.

- **Only in-plane deflections are measured.** A refractive-index gradient along the optical axis ($\partial n/\partial z$, i.e. along the camera axis) does not deflect the ray laterally and therefore produces no signal. BOS is sensitive only to gradients in the plane perpendicular to the optical axis.

- **The measurement is a gradient, not a direct surface.** The correlation pipeline recovers $\partial(\Delta n \cdot t)/\partial x$ and $\partial(\Delta n \cdot t)/\partial y$. The surface map is obtained by integrating these gradients. Integration accumulates errors: a spatially correlated noise floor in the gradient field can produce low-frequency artefacts in the reconstructed surface that are not present in the gradient maps. Always inspect both the gradient fields and the reconstruction when diagnosing anomalies.

- **Small-angle (paraxial) approximation.** The deflection-angle formula $\varepsilon \approx \tan \varepsilon$ is used throughout. For typical BOS setups with small Z_d/Z_a ratios and moderate sample-induced deflections, the error is negligible. For large deflections or extreme geometry, the approximation breaks down and a full ray-tracing model is required.

- **The reference image is assumed deflection-free.** Any structured distortion present in the reference image (lens distortion, ambient air turbulence, vibration during acquisition) will appear as a systematic offset in every result derived from that reference. Lens distortion should be corrected or characterised separately; the reference image should be acquired under the same thermal and vibration conditions as the sample images.

- **The gradient field is not guaranteed to be curl-free.** A physically consistent surface has $\partial u/\partial y = \partial v/\partial x$ everywhere. Measurement noise, outliers, and genuine optical effects (e.g. diffraction at edges) produce a curl component. The sparse Poisson solver discards this component implicitly by finding the least-squares integrable surface. The discarded curl energy is not reported, so a high-curl field can reconstruct to a smooth-looking surface while hiding significant measurement inconsistency.

10.2 Software and Implementation Caveats

- **Frankot–Chellappa remains an option, but its mask treatment is weaker.** The default reconstruction path is the masked sparse Poisson solver. Frankot–Chellappa can still be selected in the UI, but in the current implementation the mask is enforced by zeroing the gradient field outside the sample before the FFT solve. This can introduce boundary-driven artifacts that the sparse masked-domain solver avoids.

- **Reconstructed surface is relative.** The constant-offset degree of freedom is resolved by a gauge condition on one interior node. The absolute offset has no physical meaning.

- **Field correction is applied before the "raw" correlation view.** The stored raw correlation field already has the Snell's-law correction subtracted when that option is enabled. The "raw" label in the UI refers to pixel-unit (unscaled) data, not uncorrected data.

- **Validation criteria are not uniformly applied across all code paths.** The main post-processing path uses both S2N and UNO thresholding. Some alternative overloads apply only the S2N pass.
- **Reconstruction progress granularity.** Each full sparse solve is counted as one progress step. For small masks or small grids, the progress bar may jump rather than advance smoothly.
- **Geometry calibration is critical.** An incorrectly set Z_a shifts both the spatial axis and the surface amplitude by the same factor. Always calibrate against a known physical target before interpreting absolute values.

11 Conclusion

Sand Dune is a complete BOS measurement system for mapping weak optical non-uniformities in transparent media at high spatial resolution. Background pattern design, optical layout, calibration, cross-correlation, validation, and reconstruction all contribute directly to the quality of the final result. The current software combines FFT-based windowed cross-correlation, CMM sub-pixel refinement (with optional PID window deformation), UNO-based vector validation, and a selectable reconstruction stage with masked sparse Poisson integration as the default and Frankot–Chellappa as an alternative option.

At present, the most meaningful precision figure is the best-fit-plane RMS obtained from the measured `if_0.1` dataset on the central crop: approximately 0.049 px in u and 0.046 px in v with PID enabled. These figures provide a quantitative development-verification benchmark for the current implementation and establish the noise floor used elsewhere in this document. The next additions to the report should be the final repeatability figures and the completed geometry-calibration results so that the link between the hardware configuration and the measured absolute scales can be documented fully.

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